Downloading and using the phyphox magnetometer

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Australian Magnetometers-in-Schools **Class 2** (Measuring Earth's Magnetic Field with a Smart Device) and **Class 7** (Magnetism in Rocks) both use the phyphox mobile app to collect geomagnetic data. The following document contains instructions on how to download and use the phyphox app for the classes in this project.

What is phyphox?

Phyphox (short for Physical Phone Experiments) is a free app that allows you to use your smartphone or tablet for science experiments. It was developed by RWTH Aachen University in Germany. The app collects data from the sensors built into your device (such as the accelerometer, gyroscope and microphone) to collect and present real-time data from the world around you (RWTH Aachen University).

Downloading the phyphox app

The phyphox app is available for free from the Apple App Store and Google Play. The app's icon is shown below:

phyphox logo, RWTH Aachen University, CC BY 4.0.

Recording magnetic data using phyphox

When you open the app, navigate to the **Magnetometer** option (listed under Raw Sensors) (see **Figure 1**).

You will be greeted with a screen that has three graphs – one for each component of the magnetic field (x, y and z) (see **Figure 2**). This is the default **Graph** mode.

There are different modes for how to view the data. These are located in the grey taskbar at the top of the screen (see **Figure 2**):

- **Graph:** view all three components of the magnetic field (x, y and z) on separate graphs.
- **Absolute:** view the absolute (total) magnetic field strength on a single graph.
- **Multi:** view x, y, z and absolute magnetic field strength on the same graph.
- **Simple:** view x, y, z and absolute magnetic field strength numerically (no graphs).

To start recording, press the play button  in the orange taskbar at the top right of the screen (see **Figure 2**). Live readings of magnetic field strength for each component will begin to be displayed on your screen.

To stop recording, press the pause button .

You can clear the data using the delete button .

The three dots  in the top right corner of the screen opens a menu where you have options to share, save and export your data.

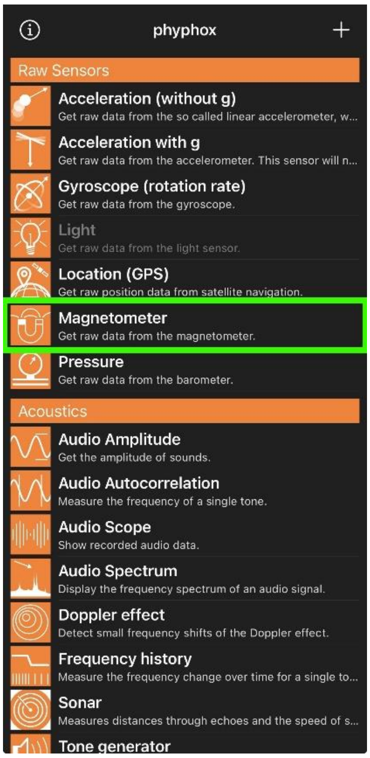

Figure 1: Labelled screenshot of the phyphox opening screen. The green box shows the magnetometer sensor. (phyphox app (RWTH Aachen University), used under CC BY 4.0)

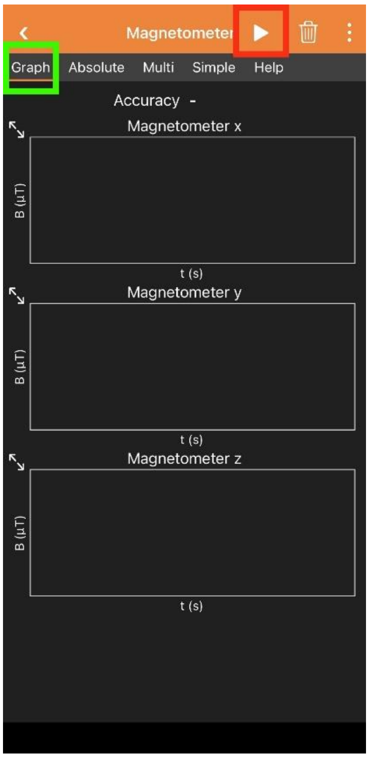

Figure 2: Labelled screenshot of the phyphox magnetometer in the default **Graph** mode (green box). Press the play button (red box) to begin recording data. (phyphox app (RWTH Aachen University), used under CC BY 4.0)

The phyphox coordinate system

If using phyphox on a mobile phone, the 3 axes are as follows:

- The z axis is perpendicular to the screen
- The y axis runs along the long edge of the phone (up is positive)
- The x axis runs along the short edge of the phone (right is positive)

More information is available from phyphox here: https://phyphox.org/sensors/

Recalibrating your device's magnetometer

When your device is exposed to large magnetic fields, the magnetometer sensor in your device can become saturated and may need recalibrating.

To do this, move your device in a figure 8 motion in all 3 directions.

The following YouTube video demonstrates how to do this: https://www.youtube.com/watch?v=-g0KXme8ISk&t=215s

References:

RWTH Aachen University. *phyphox – Physical Phone Experiments*. https://phyphox.org

RWTH Aachen University. *Sensors*. Phyphox. Available at: https://phyphox.org/sensors/